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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

VT2301-1- ADVANCED SEMICONDUCTOR MICROFABRICATION TECHNOLOGY
(For VT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In the microfabrication process, what does the term "cleanroom" refer to?
A) A room with high humidity to prevent static electricity
B) A room with controlled temperature for only cleaning processes
C) A highly controlled environment with minimal particles to prevent contamination during fabrication
D) A room for storing raw materials
2. Moor's law relates to
A) Speed of operation of bipolar devices
B) Speed of operation MOS devices
C) The power rating of MOS devices
D) Level of integration of MOS devices
3. Which of the following correctly lists the flow of semiconductor product development?
A) Packaging → Fabrication → Design → Product → System Manufacturing
B) Design → Fabrication → Packaging → System Manufacturing → Product
C) System → Fabrication → Product → Packaging → Design
D) Fabrication → Design → Packaging → Marketing → System Manufacturing
4. Which of the following is not a characteristic of miniaturization?
A) Low cost
B) Better speed of operation
C) Lower energy requirements
D) Low level of integration
5. Silicon has a _____ crystal structure
A) Simple cubic
B) Face centered cubic
C) Body centered cubic
D) Diamond
6. Which method is used for Cu deposition?
A) Thermal oxidation
B) Evaporation
C) Sputtering
D) Electroplating
7. Why silicon dioxide (SiO₂) layer is used in ICs?
A) To protect the surface of the chip from external contaminants and to allow for selective formation of the n and p regions by diffusion
B) Because it facilitates the penetration of the desired impurity by diffusion
C) To control the concentration of the diffused impurities
D) because of its high heat conduction
8. What does the "resolution" of a photoresist refer to?
A) The speed of the photolithographic process
B) The minimum feature size that can be accurately patterned
C) The thickness of the resist film
D) The color of the exposed resist

9. Which statement given below is correct with respect to Ion implantation process.
- A) Ion Implantation damages the silicon wafer surface, but that damage can be recovered by annealing process
 B) Ion Implantation damages the silicon wafer surface and that damage cannot be recovered
 C) Ion Implantation does not damage the silicon wafer surface
 D) Ion Implantation damages the silicon wafer surface, and annealing process damages it still further.
10. Which of the following is not a step involved in lithography
- A) Pre bake
 B) Photoresist application
 C) Development
 D) Evaporation
11. Which implant is used to prevent subsurface punch through between the source and drain in the transistor
- A) Well implant
 B) Halo
 C) V_T implant
 D) S/D implant
12. Which of the following is the primary goal of the PVD process?
- A) To create thin, smooth films on substrates
 B) To perform surface cleaning
 C) To deposit a layer of photoresist
 D) To etch patterns into a surface
13. To grow the polysilicon gate layer, which of the following chemical is used for chemical vapour deposition?
- A) Silicon Nitride (Si_3N_4)
 B) Silane gas (SiH_4)
 C) Silicon oxide
 D) None of the mentioned
14. Pre-deposition and Drive in are the steps involved in.....
- A) Ion implantation
 B) Thermal Oxidation
 C) PVD
 D) Diffusion
15. Which of the following is not a low-k dielectric material?
- A) SiOC
 B) SiOF
 C) Si_3N_4
 D) Porous SiOC
16. Which of the following is a High-K dielectric commonly used in modern semiconductor technology?
- A) Hafnium oxide (HfO_2)
 B) Aluminum oxide (Al_2O_3)
 C) Silicon nitride (Si_3N_4)
 D) Silicon dioxide (SiO_2)
17. For plasma etching of Silicon, which gas is generally used
- A) CH_4
 B) CO_2
 C) CO
 D) CF_4
18. The electromigration effect in interconnects leads to:
- A) Enhanced conductivity
 B) Metal void formation and circuit failure
 C) Reduced thermal stress
 D) Decreased power dissipation
19. Which of the following materials is commonly used as a diffusion barrier in semiconductor manufacturing?
- A) Silicon dioxide (SiO_2)
 B) Titanium nitride (TiN)
 C) Aluminum (Al)
 D) Copper (Cu)
20. Which of the following metals was used primarily in interconnects before the introduction of copper?
- A) Gold
 B) Silver
 C) Iron
 D) Aluminum

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Differentiate between design and fabrication in semiconductor manufacturing.	4	L*2	1	1
b)	What is system manufacturing? How does it relate to packaging and product? Elaborate using an example.	4	L2	1	1

- c) Describe the concept of optical lithography in semiconductor manufacturing. Explain the complete process flow with suitable illustrations. 8 L2 2 1
2. a) Explain the impact of packaging on the performance and reliability of an integrated circuit. 4 L2 1 1
- b) List the functions of silicon dioxide layer in IC fabrication. Also explain the thermal oxidation process with necessary equations. 4 L2 2 1
- c) With the help of neat diagrams, explain the CZ process used for silicon crystal growth. Also, describe the subsequent steps involved in preparing silicon wafers from the ingot. 8 L2 2 1
3. a) Explain how the cost-performance tradeoff for microelectronics differs from that of other industries. 4 L2 1 1
- b) Explain the significance of the CMP process in IC fabrication with the help of neat diagrams. 4 L2 2 1
- c) Explain the three important etch parameters. Differentiate isotropic and anisotropic etching. Explain in detail the wet etching process, also list its advantages and disadvantages. 8 L2 2 1

Unit – II

4. a) Describe the complete fabrication steps involved in the twin-tub CMOS process. 10 L2 4 1
- b) Discuss the role and fabrication method of salicides in deep submicron CMOS technology. Illustrate your answer with sketches. 6 L3 3 1
5. a) Explain the replacement metal gate process for gate module in detail with the help of neat diagrams. 10 L2 3 1
- b) Explain the architectural differences between planar MOSFETs and FinFETs using neat diagrams. Discuss why FinFETs are preferred in advanced technology nodes. 6 L3 4 1
6. a) What do you understand by dielectric isolation in IC fabrication? Explain the LOCOS technique in detail, including appropriate sketches, and discuss its drawbacks. 8 L2 4 1
- b) Illustrate and explain the working principle of Atomic Layer Deposition for depositing dielectrics. Why is ALD preferred for high-quality thin films? 8 L2 3 1

Unit – III

7. a) With neat diagrams, explain the Damascene technique for copper metallization. 8 L2 5 1
- b) What is electromigration? Why is it a major reliability concern in interconnects? What are the methods to reduce electromigration? 8 L2 5 1
8. a) Discuss the major integration challenges associated with Low-K dielectrics. 8 L2 5 1
- b) Describe the evolution of dielectric materials from SiO₂ to porous SiOC. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
